TEMCO LEARN

T3000 API & Protocol

JSON Schema — screenDefinition.json · built Nov 13, 2026

JSON Schema — screenDefinition.json

The complete specification for the multi-page LCD screen definition file.

Top-Level Structure

JSON

```
{
  "device": "tstat11",
  "version": "1.0.0",
  "lcdSize": { "width": 320, "height": 480 },
  "grid": { "cols": 17, "rows": 10 },
  "globalStyles": {
    "bg": "#003366",
    "fontFamily": "'Fira Mono', 'Consolas', 'Menlo', monospace",
    "fontSize": "24px",
    "fontWeight": "700",
    "textColor": "#ffffff"
  },
  "data": {
    "temp": 23.3,
    "stp": 15.5,
    "hum": 45,
    "fan": "AUTO",
    "sys": "AUTO",
    "modbus": 133,
    "baud": 115200,
    "mode": "MODBUS",
    "ip": "168.0.151"
  },
  "pages": [ ... ],
  "navigation": {
    "type": "linear",
    "order": ["main", "setpoint", "inside", "comm_settings"],
    "wrapAround": true
  }
}
```

Field Descriptions

Root Object

Field	Type	Required	Description
`device`	string	Yes	Device model identifier ("tstat10", "tstat11")
`version`	string	Yes	Schema version
`lcdSize`	object	Yes	LCD pixel dimensions
`lcdSize.width`	number	Yes	Always 320
`lcdSize.height`	number	Yes	Always 480
`grid`	object	Yes	Character grid dimensions
`grid.cols`	number	Yes	Number of columns (17)
`grid.rows`	number	Yes	Number of rows (10)
`globalStyles`	object	No	Default styles for all pages
`data`	object	Yes	Current device data values
`pages`	array	Yes	Array of page definitions
`navigation`	object	Yes	How pages are connected

`data` Object

Runtime values from the device. Each field can be referenced by widgets via the `field` property.

Field	Type	Description	Register
`temp`	number	Current temperature (°C)	Read-only sensor
`stp`	number	Setpoint temperature (°C)	Read/write
`hum`	number	Humidity (%)	Read-only sensor
`fan`	string/number	Fan mode	Read/write
`sys`	string/number	System mode	Read/write
`modbus`	number	Modbus address	Read/write
`baud`	number	Baud rate	Read/write
`mode`	string	Protocol (MODBUS/BACNET)	Read/write
`ip`	string	IP address	Read-only

`pages` Array

Each page object:

JSON

```
{
  "id": "main",
  "label": "Home",
  "icon": "house",
  "styles": {
    "bg": "#003366",
    "highlight": "#008080",
    "fontFamily": "'Fira Mono', monospace",
    "fontSize": "24px",
    "fontWeight": "700",
    "textWidthChars": 6,
    "valueBoxWidthChars": 8
  },
  "widgets": [ ... ]
}
```

Field	Type	Required	Description
`id`	string	Yes	Unique page identifier
`label`	string	Yes	Human-readable page name
`icon`	string	No	Icon for page in icon bar navigation
`styles`	object	No	Page-level style overrides (merges with globalStyles)
`styles.bg`	string	No	Background color
`styles.highlight`	string	No	Focused row highlight color
`styles.fontFamily`	string	No	Font family
`styles.fontSize`	string	No	Base font size
`styles.fontWeight`	string	No	Font weight
`styles.textWidthChars`	number	No	Label column width in characters
`styles.valueBoxWidthChars`	number	No	Value box width in characters
`widgets`	array	Yes	Array of widget objects on this page

`navigation` Object

JSON

```
{
  "type": "linear",
  "order": ["main", "setpoint", "inside", "comm_settings"],
  "wrapAround": true
}
```

Field	Type	Required	Description
`type`	string	Yes	"linear" (cycle through) or "tree" (hierarchical, future)
`order`	array	Yes	Page IDs in navigation order
`wrapAround`	boolean	No	Wrap from last to first page? Default true

Complete Example

JSON

```
{
  "device": "tstat11",
  "version": "1.0.0",
  "lcdSize": { "width": 320, "height": 480 },
  "grid": { "cols": 17, "rows": 10 },
  "globalStyles": {
    "bg": "#003366",
    "fontFamily": "'Fira Mono', 'Consolas', 'Menlo', monospace",
    "fontSize": "24px",
    "fontWeight": "700",
    "textColor": "#ffffff"
  },
  "data": {
    "temp": 23.3,
    "stp": 15.5,
    "hum": 45,
    "fan": "AUTO",
    "sys": "AUTO",
    "modbus": 133,
    "baud": 115200,
    "mode": "MODBUS",
    "ip": "168.0.151"
  },
  "pages": [
    {
      "id": "main",
      "label": "Home",
      "icon": "house",
      "widgets": [
        { "type": "wifi_icon", "row": 0, "col": 16 },
        { "type": "large_text", "field": "temp", "row": 1, "colSpan": 17, "rowSpan": 2,
        { "type": "divider", "row": 3 },
        { "type": "label_value", "id": "stp", "label": "STP", "field": "stp", "row": 3,
        { "type": "label_value", "id": "fan", "label": "FAN", "field": "fan", "row": 4,
        { "type": "label_value", "id": "sys", "label": "SYS", "field": "sys", "row": 5,
        { "type": "text", "field": "ip", "row": 6, "fontSize": "18px" },
        { "type": "icon_bar", "row": 9, "icons": [
          { "icon": "moon" },
```

```
        { "icon": "house", "pageId": "main" },
        { "icon": "snowflake", "pageId": "inside" },
        { "icon": "fan" }
      ]}
    ]
  },
  {
    "id": "setpoint",
    "label": "Setpoint",
    "widgets": [
      { "type": "wifi_icon", "row": 0, "col": 16 },
      { "type": "header", "text": "Setpoint", "row": 1 },
      { "type": "edit_value", "field": "stp", "row": 2, "fontSize": "36px", "editable"
      { "type": "icon", "icon": "house", "row": 4, "col": 1 },
      { "type": "text", "text": "Inside", "row": 4, "col": 8 },
      { "type": "large_text", "field": "temp", "row": 5, "rowSpan": 2, "fontSize": "48
      { "type": "text", "text": "HUM", "row": 8, "col": 0 },
      { "type": "text", "field": "hum", "row": 8, "col": 5, "suffix": "%" }
    ]
  },
  {
    "id": "inside",
    "label": "Inside",
    "icon": "house",
    "widgets": [
      { "type": "wifi_icon", "row": 0, "col": 16 },
      { "type": "icon", "icon": "house", "row": 1, "col": 1 },
      { "type": "text", "text": "Inside", "row": 1, "col": 8 },
      { "type": "large_text", "field": "temp", "row": 3, "colSpan": 17, "rowSpan": 2,
      { "type": "text", "text": "HUM", "row": 7, "col": 0 },
      { "type": "text", "field": "hum", "row": 7, "col": 5, "suffix": "%" },
      { "type": "icon", "icon": "snowflake", "row": 7, "col": 14, "size": "28px" },
      { "type": "icon", "icon": "fan", "row": 8, "col": 14, "size": "28px" }
    ]
  },
  {
    "id": "comm_settings",
    "label": "Comm Settings",
    "styles": {
      "highlight": "#008080",
      "textWidthChars": 6,
```

```
        "valueBoxWidthChars": 8
      },
      "widgets": [
        { "type": "header", "text": "Communication\nSettings", "row": 1 },
        { "type": "label_value", "id": "modbus", "label": "Modbus", "field": "modbus", "
        { "type": "label_value", "id": "baud", "label": "Baud", "field": "baud", "regist
        { "type": "label_value", "id": "mode", "label": "Mode", "field": "mode", "regist
        { "type": "footer_hint", "text": "+  Edit  -", "row": 8 },
        { "type": "footer_nav", "text": "< Back   Next >", "row": 9 }
      ]
    }
  ],
  "navigation": {
    "type": "linear",
    "order": ["main", "setpoint", "inside", "comm_settings"],
    "wrapAround": true
  }
}
```

Validation Rules

1. Every page `id` must be unique
2. Every page `id` referenced in `navigation.order` must exist in `pages`
3. Widget `field` values must reference keys in the `data` object
4. Widget `register` values must be valid Modbus register numbers
5. `options` arrays with all numbers should be sorted ascending
6. `row` values must be 0 to `grid.rows - 1`
7. `col` values must be 0 to `grid.cols - 1`